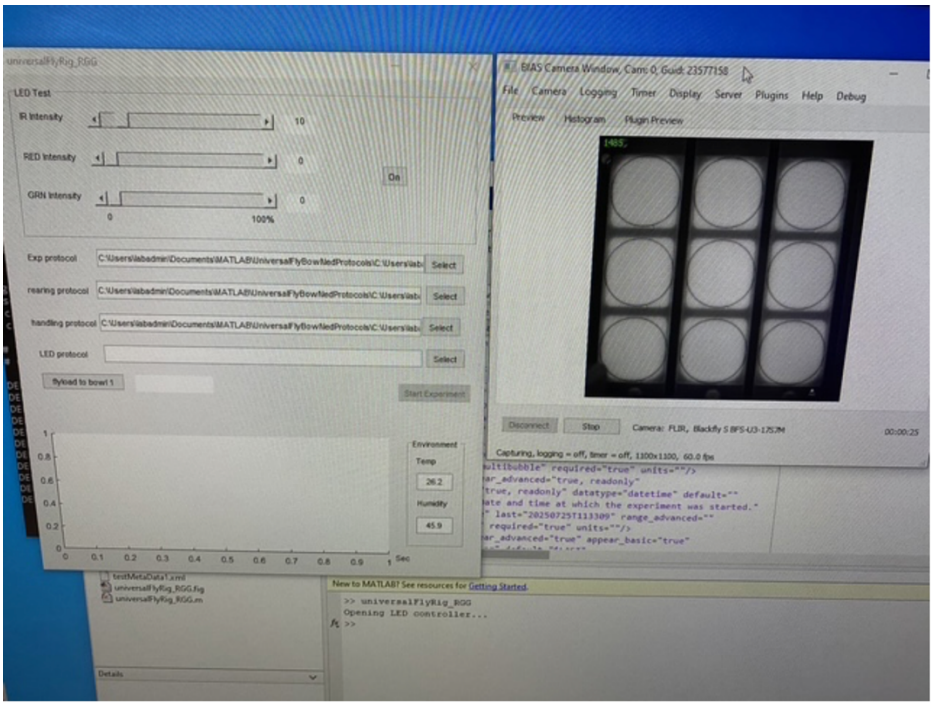
Quick Guide on Running Universal Behavior Rigs

1. Turn on power strip for overhead lights, cameras, and chiller. For overhead lights, check that they are set to the right level. You can change this by pressing the on off switch on the strip inside the rig

2. To start software: Start MATLAB. For setups with RGB lightboard, run **flyBowl_RGB** in the MATLAB command window. For setups with RGG lightboard, run **universalFlyRig_RGG** in the MATLAB command window. A GUI and BIAS camera window should appear, as below:



3. Load in your LED protocol in GUI. Click the select button to the right of the LED protocol field and navigate to desired LED protocol. Optionally, you may also load experimental, handling, and rearing protocols. If loaded properly, a time series of LED on/off states should appear in GUI.

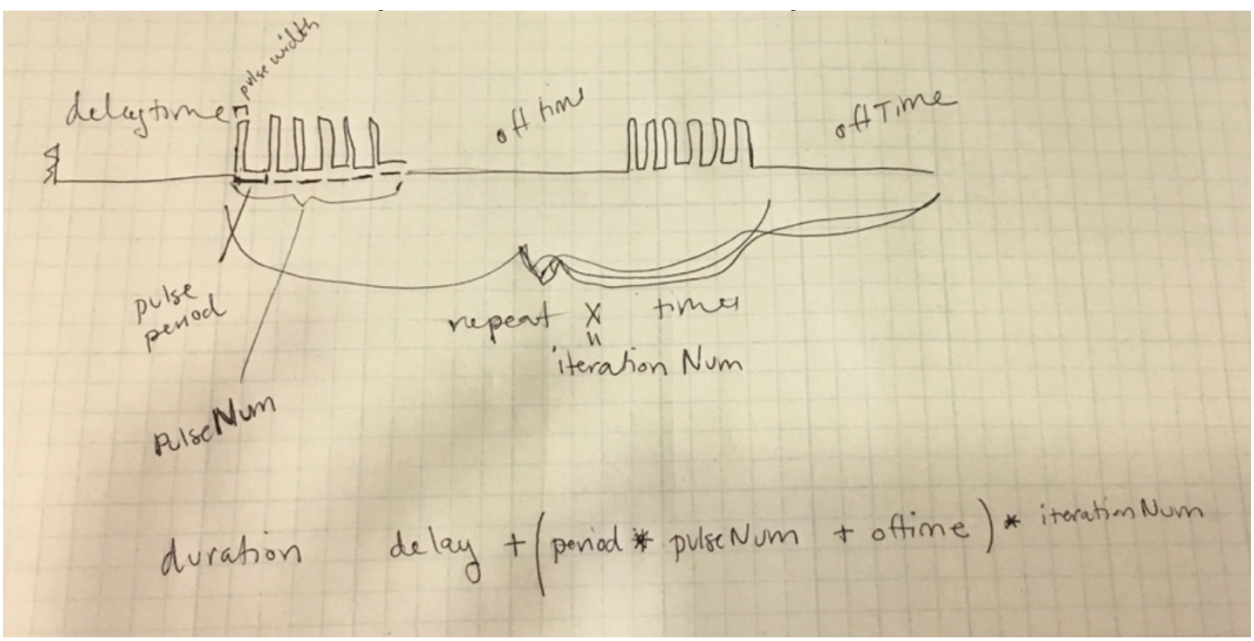
4. Once loaded with flies, place behavioral chambers on the setup and click "Flyload to bowl 1." This will provide a timestamp for when flies were placed on system and calculation of time between chamber loading and experiment start.

5. Once ready, click Start Experiment on lower right. Upon experiment start, a second GUI should appear for metadata entry. The following fields* should be entered

(at minimum): username, screen_type, sex(es), male_parent (genotype), female_parent (genotype), driver, effector, number of flies, number flies dead (if any). If need to stop the experiment, click “ABORT” (which will appear in prior location of Start Experiment button). A flag will be generated in the resulting experiment directory.

* Default values or strings in the Metadata GUI dropdown menus may be altered in the flybowlMetaTree_multibubble.xml file situated in the UniversalRig_RGG or UniversalRig_RGB parent directory.

6. Once experiment is complete, close the metadata GUI window (this initiates writing of experimental files).

7. Once all experiments in the session are complete, close out of the GUIs and MATLAB. Shut power strip off for system.

<u>Additional Notes on Designing LED protocols</u>